

08/08/2026

C03-AT2

1) A shepherd boy gets bored tending the town's flock. To have some fun he calls out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy ~~saw~~ a real wolf approaching the flock and calls out. The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes into panic. Identify TP, FP, FN, TN.

✓

TP

Real wolf
came warning
also given

N

FN

wolf came
but warning is not ~~given~~

N

FP

No wolf came
false warning
given

TN

No wolf
No warning

2.) Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and calculate the Precision and Recall and F1 score.

Total Images = 100

Total cat image = 30

TP = 25, FN = 5

FP = 5, TN = 55

30, 70, 100

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{25}{25 + 5} = \frac{25}{30} = 0.83 = 83\%$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{25}{25 + 5} = \frac{25}{30} = 0.83 = 83\%$$

$$= 0.55 \approx 55\%$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 83.3 \times 55.5}{83.3 + 55.5}$$

$$= 68.5$$

3) All email service provider want to classify emails as spam or No spam using features.

- * contains "offer" (Yes/No)
- * contains "Free" (Yes/No)
- * Email Length (short/Long)

offer	Free	length	class
Yes	Yes	short	spam
Yes	No	Long	Not spam
No	Yes	short	Spam
No	No	long	Not spam
Yes	Yes	Long	spam
No	No	short	No spam

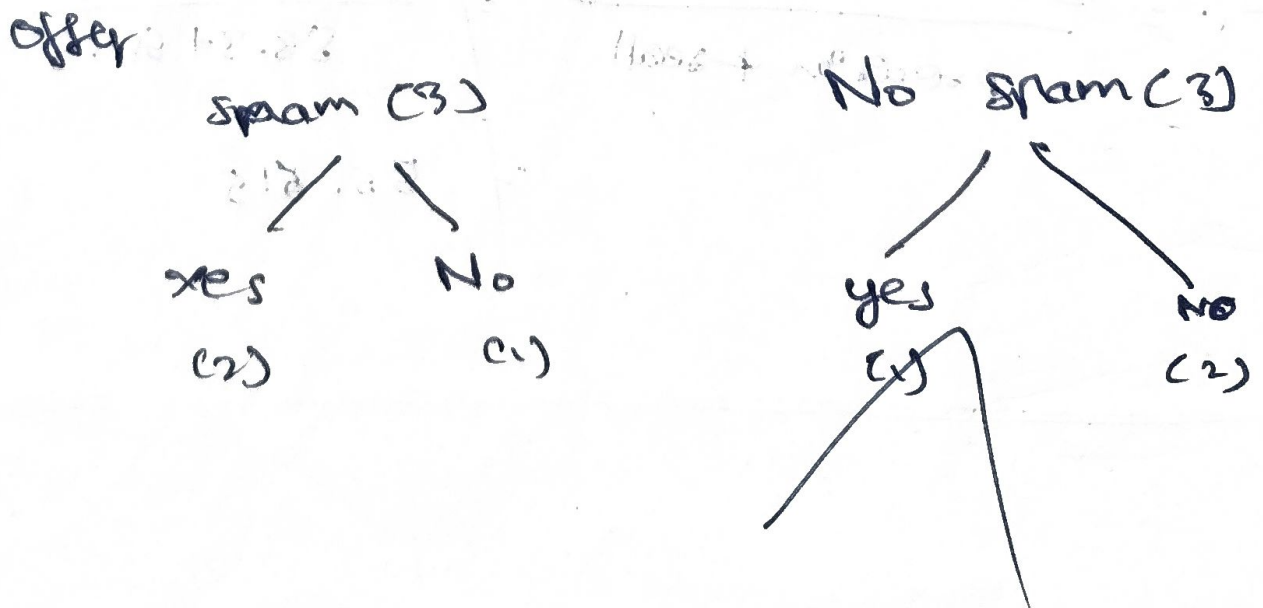
1. Prior probabilities

Total Spams = 6

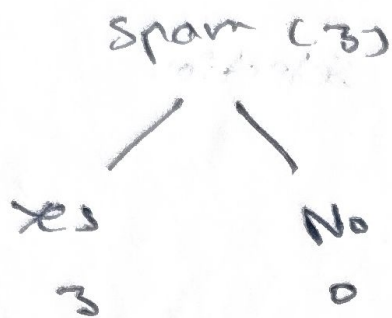
$$\text{Spam} = 3 = \frac{3}{6} \quad P(\text{spam}) = \frac{3}{6}$$

$$\text{Not spam} = 3 = \frac{3}{6} \quad P(\text{not spam}) = \frac{3}{6}$$

2) Conditional probabilities



Free



length

spam (3)



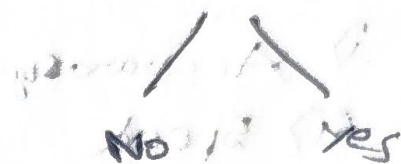
$$P = 2/3$$

$$P = 1/3$$

$$P(\text{Free} = \text{yes} / \text{spam}) = 3/3 = 1$$

$$P(\text{Free} = \text{No} / \text{spam}) = 0/3 = 0$$

No spam (3)



No spam (3)



$$P = 1/3$$

$$P = 2/3$$

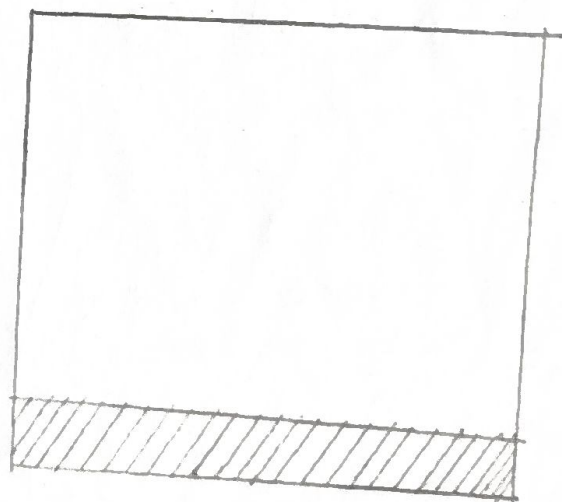
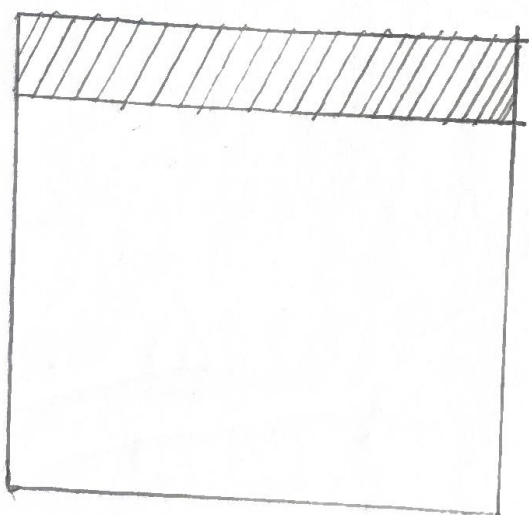
$$P(\text{Free} = \text{yes} / \text{no spam}) = 0/3 = 0$$

$$= 0/3 = 0$$

$$P(\text{Free} = \text{No} / \text{no spam}) = 3/3 = 1$$

$$= 3/3 = 1$$

Graph:



Khani